

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		helium, argon, xenon ☐ helium, neon, argon ☐ argon, krypton, xenon ☐ krypton, xenon, radon ☒			1	1		
		(ii)		XeF ₂ ☒ KrF ₂ ☐ XeCl ₂ ☐ KrCl ₂ ☐			1	1		
		(iii)		23 °C ☒ 71 °C ☐ 110 °C ☐			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		award (1) for description of smaller atom neon has fewer electron shells neon is a smaller atom (than argon / krypton / xenon) neon's outer electrons are closer to the nucleus (than in argon / krypton / xenon) neon has an atomic radius < 80 pm award (1) for effect of small size less shielding (than argon / krypton / xenon) stronger attraction between outer electron and nucleus (than in argon / krypton / xenon) outer electrons are more difficult to remove (than in argon / krypton / xenon)			2	2		
	(b)			award (1) for use and (1) for relevant linked property e.g. - weather balloons / airships / party balloons ⇒ helium has very low density / lower density than air - underwater breathing apparatus ⇒ helium is inert - MRI scanners ⇒ helium has a very low boiling point accept other sensible answers e.g. airships ⇒ helium is inert	2			2		
				Question 7 total	2	0	5	7	0	0